PROJECT REPORT

NAME: ARNAV SINGH
REG NO. : 25BAI11364
SUBJECT CODE: FUNDAMENTALS IN AI AND ML (CSA2001)
BRANCH: COMPUTER SCIENCE & ENGINEERING (AIML)
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TITLE

AI-Based Rock-Paper-Scissors Game Using CustomTkinter

This project is an intelligent version of the traditional Rock-Paper-Scissors game developed using Python. It integrates basic Artificial Intelligence techniques and a graphical user interface using CustomTkinter to create a more interactive and adaptive gameplay experience compared to standard random-based implementations.

INTRODUCTION

The project focuses on developing a smart game system where the computer opponent can predict user moves based on past behavior. Unlike traditional implementations, this system uses pattern recognition and frequency analysis. The graphical interface enhances usability, making the application visually appealing and easy to interact with for users.

MOTIVATION OF THE PROJECT

Traditional games lack intelligence and become repetitive over time. The motivation behind this project is to introduce adaptive gameplay where the system learns from user actions. It helps demonstrate basic AI concepts in a simple and understandable way while making the game more challenging and engaging.

PROBLEM STATEMENT

Existing Rock-Paper-Scissors games rely on random decision-making, making them predictable and less engaging. The problem is to develop a system where the computer can learn from previous moves and make intelligent decisions, thereby improving gameplay and demonstrating real-world AI concepts.

OBJECTIVE OF THE PROJECT

The main objective is to design and develop a GUI-based game that integrates AI techniques. The system aims to predict user moves, provide a better gaming experience, and demonstrate learning behavior. It also focuses on improving user interaction through an intuitive and responsive graphical interface.

EXISTING METHODS

Traditional implementations of this game rely on random number generation or fixed rules. These methods do not adapt to user behavior and lack learning capabilities. As a result, they fail to provide a dynamic experience and do not demonstrate any form of intelligent decision-making

PROS AND CONS OF THE STATED METHODS

Advantages include simplicity, low computational cost, and ease of implementation. However, the disadvantages are significant as these systems lack intelligence, cannot learn from past data, and become predictable. This reduces user interest and fails to simulate real-world intelligent systems.

HARDWARE AND SOFTWARE REQUIREMENTS

The project requires basic hardware such as a computer or laptop with at least 4GB RAM. Software requirements include Python programming language, CustomTkinter library for GUI, Tkinter support, and an IDE such as Visual Studio Code or PyCharm for coding and execution.

METHODOLOGY AND GOAL

The system uses a combination of pattern recognition and frequency analysis to predict user behavior. A small amount of randomness is included to avoid predictability. The goal is to create a semi-intelligent system that adapts over time and enhances the gaming experience.

FUNCTIONAL MODULES DESIGN AND ANALYSIS

The system consists of multiple modules including user input, AI prediction, data storage, game logic, and GUI display. Each module performs a specific function, ensuring smooth execution. The modular design improves maintainability, scalability, and clarity of the overall system architecture.

ALGORITHM DEVELOPMENT

The algorithm begins by taking user input and storing it in memory. It then analyzes past moves using frequency and pattern matching techniques. Based on the prediction, the AI selects a counter move. Finally, the system determines the result and updates the interface accordingly.

SOFTWARE ARCHITECTURAL DIAGRAM

The system follows a simple architecture where the user interacts with the GUI. The GUI sends input to the AI engine, which processes the data and returns the result. The output is then displayed back to the user, forming a continuous interaction loop.

CODING

The project is implemented using Python, utilizing the CustomTkinter library for GUI development. The code includes modules for AI prediction, pattern storage, and game logic. The structured and modular coding approach ensures readability, maintainability, and efficient execution of the program.

OUTPUT

The output is a graphical interface displaying the user's move, the AI's move, and the result of each round. The interface updates dynamically based on user interaction, providing immediate feedback and making the game visually engaging and easy to understand.

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

Key implementations include pattern recognition, frequency analysis, adaptive AI decision-making, and GUI integration. These features work together to create a system that learns from user behavior and responds intelligently, making the gameplay more dynamic and interesting.

SIGNIFICANT PROJECT OUTCOMES

The project successfully demonstrates how basic AI techniques can be applied to simple games. It improves user engagement, introduces adaptive behavior, and provides a better understanding of AI concepts. The GUI enhances usability and overall user experience.

TESTING AND REFINEMENT

The system was tested with multiple input patterns to evaluate AI performance. Edge cases and invalid inputs were handled to ensure robustness. Refinements were made to improve prediction accuracy and maintain a balance between intelligence and unpredictability.

PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

This project can be applied in game development, behavioral analysis, and decision-making systems. The concepts used here can be extended to more complex AI applications such as recommendation systems, predictive analytics, and human-computer interaction systems.

CONTRIBUTION / FINDINGS OF THE PROJECT

The project demonstrates that even simple AI techniques can significantly enhance user interaction. It highlights the importance of learning systems and shows how adaptive behavior can improve engagement. It also provides practical experience in integrating AI with GUI applications.

LIMITATION / CONSTRAINTS OF THE SYSTEM

The system uses basic AI techniques and does not implement advanced machine learning models. It may not perform well against highly unpredictable users. Additionally, the learning is limited to short-term patterns and does not involve deep data analysis.

CONCLUSIONS

The project successfully integrates AI with a simple game to create an intelligent and interactive system. It demonstrates the effectiveness of pattern recognition and frequency analysis in decision-making. The GUI enhances usability, making the project both educational and engaging.

FUTURE ENHANCEMENTS

Future improvements may include adding machine learning algorithms, multiplayer functionality, score tracking, and enhanced graphics. The AI can be made more sophisticated using reinforcement learning techniques, making the system more accurate and adaptable.

REFERENCES

Python Official Documentation, CustomTkinter Documentation, AI learning resources and tutorials.